

FPGA 부분 재구성을 이용한 적응적 쌍별 시퀀스 정렬 가속

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Adaptive Pairwise Sequence Alignment Acceleration via FPGA Partial Reconfiguration

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Abstract

Recent advances in sequencing technologies have caused genomic data to grow faster than conventional computing platforms can efficiently process, making pairwise sequence alignment (PSA) a persistent bottleneck in modern genome-analysis pipelines [1]-[6]. Although FPGA accelerators have shown strong performance and energy efficiency for dynamic-programming-based alignment, most existing designs remain static and are optimized for a single algorithm, fixed sequence length, or fixed scoring configuration [1], [3], [5]. In practice, different PSA workloads can require different methods; as an example, very short sequences favor Myers bit-vector methods, medium-similarity inputs benefit from banded Smith-Waterman, low-similarity cases require full Smith-Waterman, and high-similarity inputs are suitable for sparse approaches such as Wavefront Alignment. This paper proposes an adaptive architecture using partial reconfiguration (PR), where a lightweight Myers engine is retained in the static region and the reconfigurable region selectively loads Banded Smith-Waterman, full Smith-Waterman, or Wavefront Alignment according to input similarity and sequence length characteristics. The proposed design aims to reduce unnecessary hardware overhead maintains alignment accuracy while tracking the best-performing alignment regime across heterogeneous workloads. This study positions partial reconfiguration as a practical mechanism for workload-aware and energy-efficient genomic acceleration on FPGA platforms.

Keywords — *FPGA, partial reconfiguration, pairwise sequence alignment, Smith-Waterman, dynamic programming, genomics acceleration, SoC-based accelerator design*

I. Introduction

Biological sequence alignment is a fundamental operation in read mapping, assembly, and downstream genomic analysis. With the rapid growth of sequencing throughput, the amount of generated genomic data increasingly exceeds the pace of improvement in general-purpose computing systems, creating a serious performance bottleneck in practical pipelines [2], [5]. Dynamic-programming-based methods such as Smith-Waterman remain attractive because they provide accurate local alignment. However, their computational cost is substantial [1], [3].

Field-Programmable Gate Arrays (FPGAs) is a feasible platform for this workload because they provide fine-grained parallelism, customizable datapaths, and good energy efficiency. Prior surveys have shown that FPGA-based sequence-alignment accelerators can achieve large speedups over CPU implementations by exploiting systolic-array parallelism and application-specific dataflow [1], [3], [5]. However, most prior accelerators are designed for one alignment method or one operating regime, which limits their efficiency when real workloads vary in read length, similarity, and scoring requirements [1].

Our observation is that no single pairwise-alignment algorithm is optimal across all workload regimes. Myers bit-vector alignment is well suited to very short sequences, Banded Smith-Waterman offers a strong tradeoff for medium-similarity inputs, full Smith-Waterman is needed for low-similarity difficult cases, and Wavefront Alignment can be more efficient for highly similar

pairs. This motivates an adaptive hardware organization capable of switching between specialized engines at runtime.

II. Related Work

Existing surveys consistently report that FPGA-based sequence-alignment research has primarily progressed as a collection of algorithm-specific accelerators, including Smith-Waterman systolic arrays, affine-gap extensions, BLAST-inspired pipelines, and FM-index-based mappers [1], [3], [5]. These studies demonstrate strong benefits from specialization, but they also highlight a lack of adaptive and scalable architectures that can efficiently handle heterogeneous genomic datasets [1]. GenDP, a framework for accelerating dynamic-programming kernels in genome sequencing pipelines, further shows that modern genomics pipelines are increasingly composed of multiple dynamic-programming kernels with different dependency patterns, objective functions, and precision requirements [2]. This observation suggests that future accelerators should not only optimize a single kernel but also adapt to diverse computational regimes. **In parallel, ONT-X demonstrates that partial reconfiguration is practical in portable genomics systems, especially when different kernels or pipeline phases require different latency and throughput tradeoffs [4]. Despite these advances, PR has not been sufficiently explored inside the PSA stage itself.** That gap motivates this work. Instead of using reconfiguration only across coarse pipeline stages, this paper considers algorithm-level adaptation within pairwise PSA.

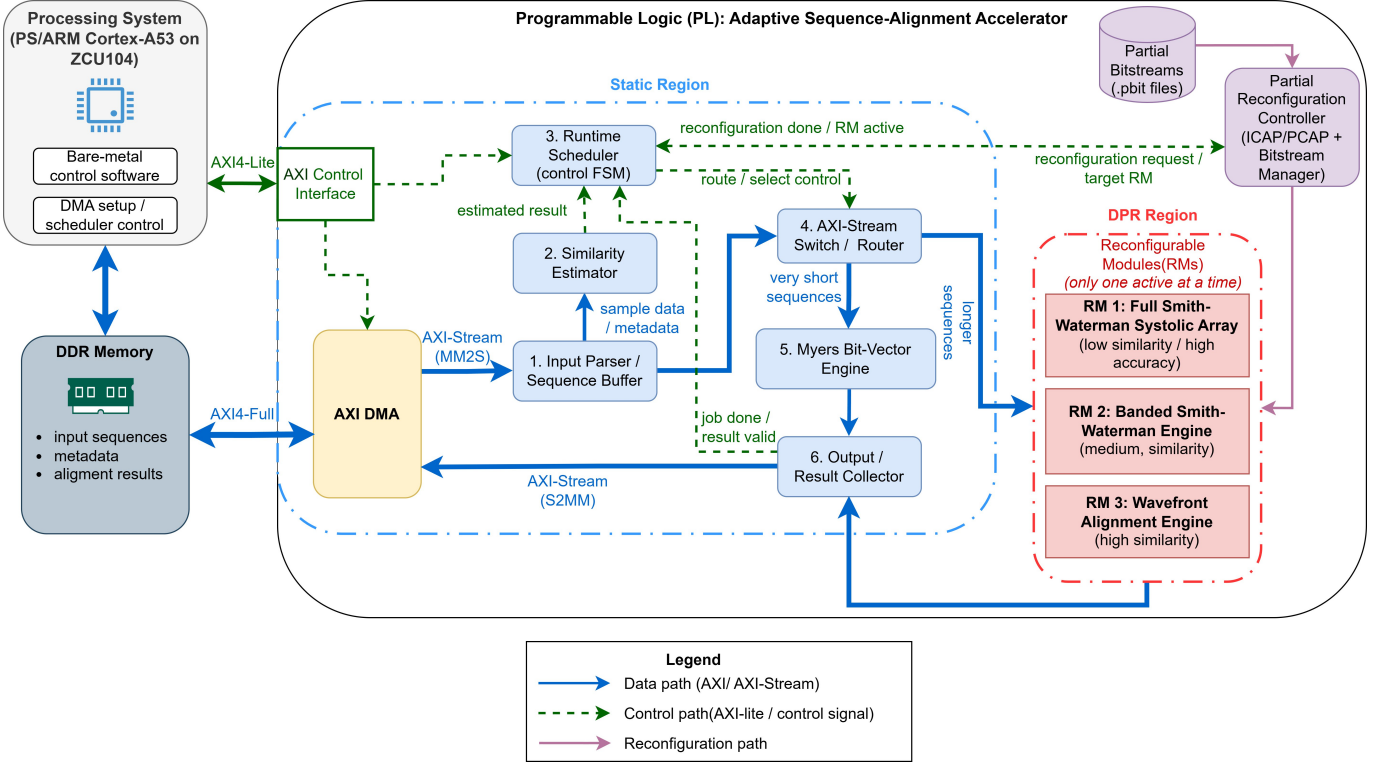


Figure 1. System architecture of the proposed adaptive sequence-alignment accelerator.

III. Proposed Architecture

The proposed architecture is divided into a static region and a dynamically reconfigurable region. The static region contains a lightweight similarity estimator, a runtime scheduler, AXI-based control and stream interfaces, and a Myers bit-vector engine for the very-short-sequence fast path. Because Myers is compact and likely to be invoked frequently, placing it in the static region avoids unnecessary reconfiguration overhead.

The reconfigurable region hosts three heavyweight alignment engines: Banded Smith-Waterman for medium-similarity inputs, full Smith-Waterman for low-similarity inputs, and Wavefront Alignment for highly similar sequence pairs. A similarity-aware scheduler selects the engine based on simple sequence features such as input length and estimated similarity. This organization aims to achieve the efficiency of specialized accelerators while preserving flexibility across diverse workloads.

At the system level, the design targets a ZCU104 platform with AXI-based communication between the processing system and programmable logic. Static and PR regions share a unified input/output format so that different alignment engines can be evaluated under a common experimental framework.

IV. Experimental Design

The evaluation will compare four classes of implementations: (1) bare-metal CPU baselines on the ZCU104 ARM cores, (2) standalone static FPGA accelerators for Smith-Waterman, Banded Smith-Waterman, and Wavefront Alignment, (3) the static Myers engine, and (4) the proposed adaptive partial-reconfiguration system. CPU baselines will include optimized Smith-Waterman

and Myers implementations, with Banded Smith-Waterman used as a mid-range software reference.

Experiments will vary sequence length and similarity to expose the algorithm-dominance regions of each method. Key metrics will include throughput in GCUPS, latency per alignment, alignment accuracy, energy efficiency, FPGA resource utilization, and the amortized overhead of PR. A key objective is to show that the proposed adaptive architecture tracks the upper performance envelope of the best static engine across mixed workloads without sacrificing correctness.

Representative figures in the final paper will include throughput-versus-similarity curves, speedup over the best static FPGA baseline, reconfiguration-overhead sensitivity analysis, and ablation studies that remove the similarity estimator or Banded Smith-Waterman to quantify the value of each architectural component.

The expected contributions of this work are as follows. First, it proposes a workload-aware FPGA architecture for PSA based on PR. Second, it introduces a practical partitioning strategy that keeps the Myers engine in the static region while reserving the PR region for heavier alignment kernels. Third, it integrates Banded Smith-Waterman, full Smith-Waterman, and Wavefront Alignment into a unified adaptive framework guided by a lightweight similarity estimator. Finally, it establishes an evaluation methodology that compares CPU, static FPGA, and PR-based designs under a common workload model.

V. Summary

This paper presents a full research plan for adaptive PSA acceleration using FPGA partial reconfiguration. The proposed

system is motivated by the mismatch between static accelerator design and the heterogeneous nature of real alignment workloads. By combining a static short-read fast path with selectively reconfigurable heavyweight engines, the architecture seeks to improve both throughput and efficiency across diverse similarity regimes. Ongoing work focuses on completing the CPU baselines, implementing the static FPGA accelerators, and integrating the final PR runtime for experimental validation.

References

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